

Carlos Emmanuel Mendiola Miranda

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Available for full-time work. I study two engineering degrees online and schedule classes outside working hours.

Education

Hybridge Education

Software Engineering and Artificial Intelligence Engineering

Online

Currently enrolled

CBT San Pedro Tlaltizapán

Tianguistenco, State of Mexico

Computer Technician

July 16, 2026

- Passed the professional examination unanimously and with honors.

Experience

FER&REN

Toluca, State of Mexico

Part-time DevOps Consultant

January 2026 – Present

- Built self-hosted high-availability infrastructure on Dokploy, Directus and PostgreSQL with Patroni and etcd, after an AWS us-east outage took all five Flutter applications down.
- Configured a three-node cluster across a VPS, a homelab reached through Cloudflare Tunnels and an iMac, so losing one node leaves two serving.
- Migrated the five Flutter applications to Astro, Preact and Tailwind CSS.
- Developed five more applications on that infrastructure: vehicle inspection, service management, personnel administration, income and expense control, and a PPF installation checklist.
- Wrote two Go services that send the scheduled notifications and report when a container stops responding.
- Added features to the ten production applications as the business needs changed.

FER&REN

Metepec, State of Mexico

Systems Developer

July 2025 – December 2025

- Developed five cross-platform Flutter applications on Supabase: attendance, PPF quoting, PPF inventory, sales records and studio income.
- Modeled the data and the business rules on the workflows the shop already followed.
- Trained the staff and handled incidents on the applications in production.

Projects & Certifications

Declarative infrastructure

Own project

Design, provisioning and operation

2026

- Provisioned an OVHcloud bare-metal server with five Terraform stacks whose state lives in Cloudflare R2, and left everything inside a guest to Ansible.
- Stood up a three-member control plane with etcd, a PostgreSQL primary with a streaming standby and Redis, keeping etcd snapshots off the server.
- Chained Jenkins, GHCR by sha256 digest, a commit to the gitops repository and Argo CD, so a push ends as an HTTPS response with no manual step.
- Put Prometheus, Loki and Grafana on a guest outside the cluster they watch, because a dashboard served by the cluster goes down exactly when it is needed.
- Left WireGuard as the only door and every secret in Bitwarden Secrets Manager, read by a single script; nothing travels on a command line.

- Tested recovery against the live cluster: promoted the PostgreSQL standby through timeline 2 and re-seeded it, then restored etcd, verified it and reverted it.

Pokédex Manager

Open source

Product design, architecture and implementation

2026

- Built the application on TanStack Start, React 19, MongoDB and Better Auth, with every collection query scoped to the session user.
- Wrote a read-only MCP server with six tools and four resources: the assistant connects over an in-memory transport and external clients over Streamable HTTP with an explicit bearer and subject.
- Wired the Kimi model into conversation, card recognition, insights and generative research, and checked every output against PokéAPI before showing it.
- Left the application standing with no AI provider: catalog, collection, statistics and a local assistant still answer through the same MCP boundary.
- Verified 26 MongoDB indexes on every start, and put GitHub Actions on audit, lint, types, tests and build against a real MongoDB.

FER&REN Detailing

ferrendetailing.com.mx

Product design and implementation

2026

- Published nineteen service routes in Astro, plus brands, about and location.
- Built the price tables from the studio documentation and left every service whose price depends on the vehicle as quote-only.
- Opened WhatsApp from each service with the message already written, so the visitor arrives saying what they came to ask.
- Loaded GSAP only on the home page; the rest of the catalog reads without JavaScript.

2 Free

Open source

Product design, architecture and implementation

2026

- Split the ecosystem into four repositories (API, web application, shared packages and landing) and published the @camircode/twofree-* packages to GitHub Packages from GitHub Actions on a version tag.
- Built the API with NestJS 11, Prisma 7 and Better Auth, bundled by esbuild into a distroless image with no shell that runs as uid 65532 on a read-only filesystem.
- Separated the image that applies migrations from the one that serves requests: the first inspects the schema before touching it and refuses to migrate a database it classifies as drifted.
- Served the Next.js 16 application behind a route handler that forwards to the in-cluster API service, which has no public address, and keeps the session cookie first-party.
- Modeled money as an integer coefficient and a scale, with no binary floating point.
- Built the Linux and Android applications with Tauri, Rust and SQLCipher: the encrypted local mode works with no account and no network.

Hybridge Education

Online

Certificates

September 4, 2026

- “Uso y dominio de Claude Code” (use and command of Claude Code).
- “Uso y dominio de Git y Github” (use and command of Git and GitHub), quoted as Hybridge issued it.

Tecnológico Nacional de México

Online MOOC

Public Center for Artificial Intelligence Training Preparatory Course

July 20, 2026

- Passed the evaluated modules in networking, cloud computing, cybersecurity, Java, artificial intelligence and data analysis.

Skills & Interests

- Method: shipped on six stacks with nothing in common (Flutter and Dart, Astro and TypeScript, Go, Rust and Tauri, NestJS, Kubernetes and Terraform), and learn the next one from its documentation with AI agents.
- Technical: TypeScript, JavaScript, Go, Rust, Python, Dart, SQL, HTML and CSS; Astro, React, Next.js, Preact, NestJS, Prisma, Tauri, Flutter and Tailwind CSS; PostgreSQL, MongoDB and Redis; Docker, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, Argo CD, Prometheus, Grafana, WireGuard, Bitwarden Secrets Manager, Linux, Bash and Git; AI agents (Claude Code, OpenCode, Codex and Pi), Model Context Protocol (MCP) and Spec-Driven Development.
- Soft: Proactivity, autonomy, leadership, problem solving, adaptability, technical language, time management and strategic thinking.
- Languages: Spanish (native) and English (A2).
- Interests: High-fidelity audio, video games, desktop environment customization, window managers and compositors, free and open source software, and personal finance.